

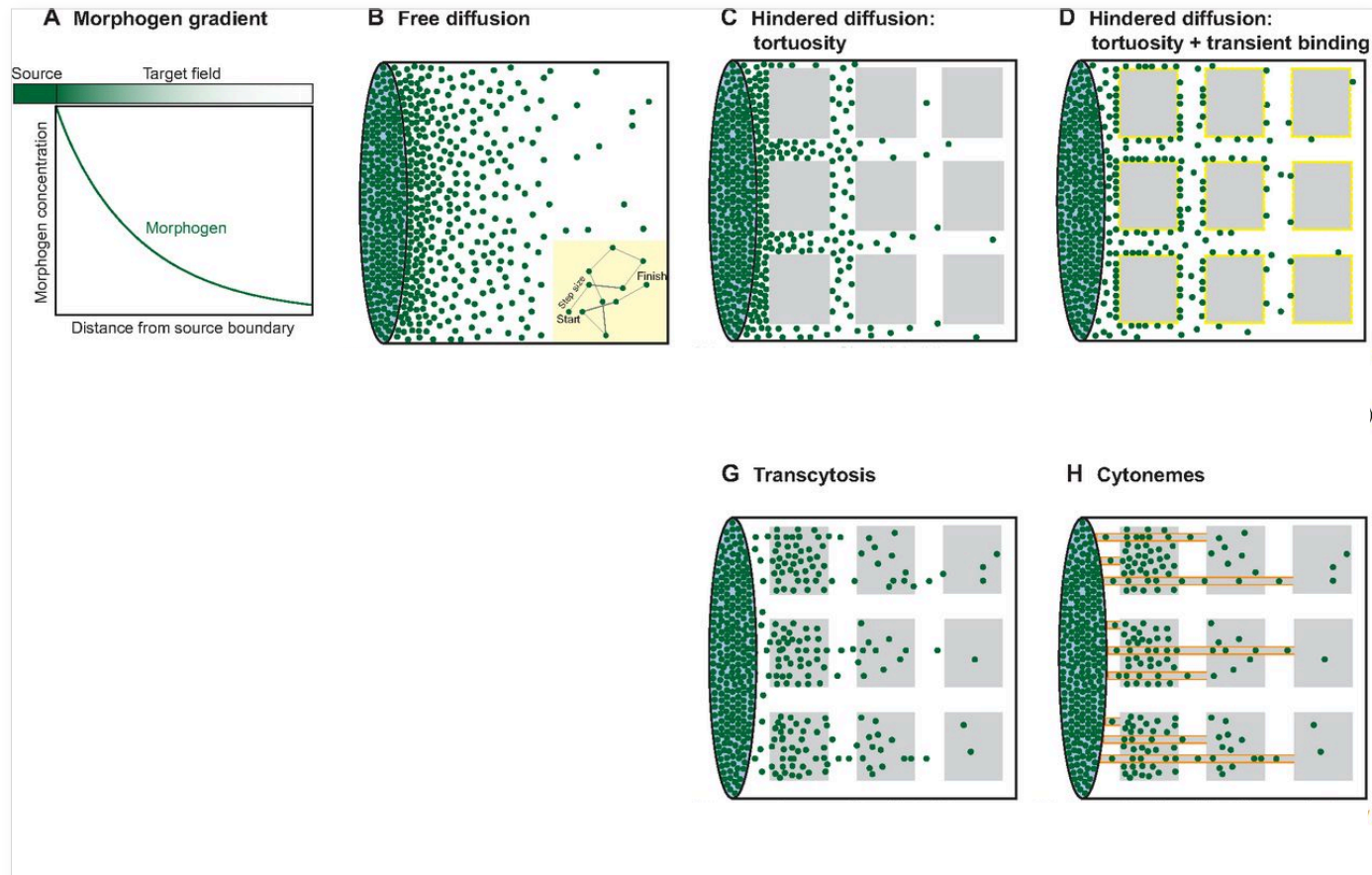
Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

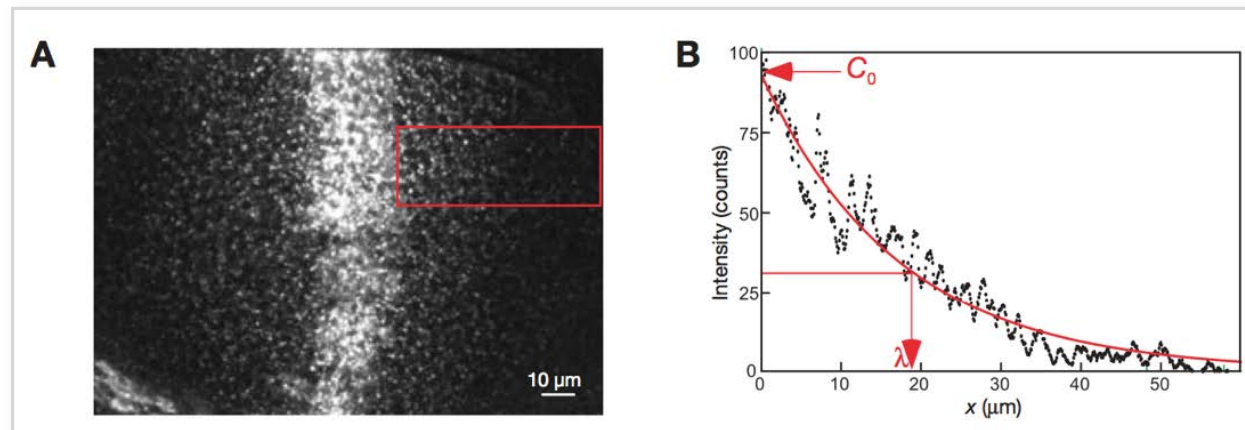
What we covered

- Quantitative imaging of morphogen gradients in *Drosophila melanogaster*
 - Diffusion or planar transcytosis
-

Morphogen transport models



Measuring morphogen gradient shape



C_0 , the gradient amplitude, is the concentration at the source boundary, and the decay length λ is the distance to the source in which the concentration has decayed to C_0/e . In steady state, λ depends on diffusion and degradation:

$$\lambda = \sqrt{D/k},$$

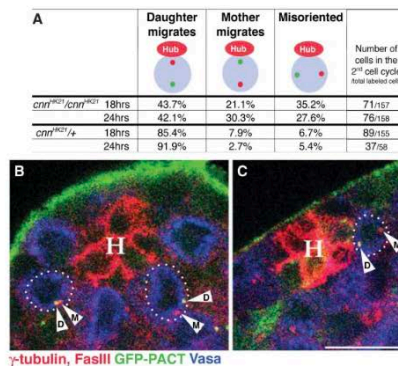


Fig. 4. *cnn* is required for nonrandom segregation of mother and daughter centrosomes. **(A)** Summary of the centrosome-positioning pattern in *cnn^{H221}* homozygous mutant and control *cnn^{H221}/+* GSCs. Daughter centrosomes were labeled by a pulse of GFP-PACT as in Fig. 1B. Only counts of cells in the second cell cycle are shown. **(B)** and **(C)** Testis tips from *cnn* males with GSCs in the second cell cycle with misoriented centrosomes (**B**) or with the mother rather than the daughter centrosome segregated to the opposite side of the GSC (**C**). Red, γ -tubulin [centrosomes are shown with arrowheads (M, mother; D, daughter)] and Fas III (hub, H); green, GFP-PACT; blue, Vasa (germ cells). Scale bar, 10 μ m.

assembled many cell generations earlier. In contrast, the daughter centrosome migrates away from the niche and is inherited by the cell that will initiate differentiation. We postulate that the mother centrosomes in male GSCs may remain anchored to the GSC-niche interface throughout the cell cycle by attachment to astral microtubules connected to the adherens junction, whereas the daughter centrosomes may initially have few associated microtubules and thus can move away from the niche. Microtubule-dependent differential segregation of mother and daughter spindle-pole bodies (equivalent to centrosomes in higher organisms) is observed in budding yeast (13). In cultured vertebrate cells, the centrioles mature slowly over the cell cycle, and the mother centrosomes (containing a mature centriole) attach astral microtubules more effectively and are more stationary than daughter centrosomes in interphase (14). The unusually early separation of centrosomes in interphase male GSCs may provide a way to move the daughter centrosome out of range of the stabilizing influence of the adherens junction complex before it becomes competent to hold a robust microtubule array.

Developmentally programmed anchoring of the mother centrosome may provide a key mechanism to ensure the stereotyped orientation of the mitotic spindle and thus the reliably asymmetric outcome of the male GSC divisions. Although it is tempting to speculate that de-

terminants associated with the mother or daughter centrosome may play a role in specifying stem cell or differentiating-cell fates, such determinants are yet to be identified. Rather, the asymmetric inheritance of mother and daughter centrosomes in male GSCs may be a consequence of the cytoskeletal mechanisms that are

Kinetics of Morphogen Gradient Formation

Anna Kicheva,^{1,2*} Periklis Pantazis,^{1,*†} Tobias Bollenbach,^{3,*†} Yannis Kalaidzidis,^{1,4} Thomas Bittig,³ Frank Jülicher,^{3,5} Marcos González-Gaitán,^{1,2,5}

In the developing fly wing, secreted morphogens such as Decapentaplegic (Dpp) and Wingless (Wg) form gradients of concentration providing positional information. Dpp forms a longer-range gradient than Wg. To understand how the range is controlled, we measured the four key kinetic parameters governing morphogen spreading: the production rate, the effective diffusion coefficient, the degradation rate, and the immobile fraction. The four parameters had different values for Dpp versus Wg. In addition, Dynamin-dependent endocytosis was required for spreading of Dpp, but not Wg. Thus, the cellular mechanisms of Dpp and Wingless spreading are different: Dpp spreading requires endocytic, intracellular trafficking.

Although the molecular and cellular mechanisms controlling morphogen transport have received much attention, many questions remain open (1–7). This might be due in part to the existence of only a few quantitative

imposed as part of the stem cell program to anchor one centrosome next to the niche throughout the interphase, ensuring a properly oriented spindle.

References and Notes

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- We thank the Bloomington Stock Center, the Developmental Studies Hybridoma Bank, and R. Lehmann for reagents; the Stanford Cell Science Imaging Facility for assistance in microscopy; Fuller lab members for comments; the Jose Carreras International Leukemia Foundation for a fellowship to Y.M.Y.; and NIH for grant P01 DK53074 to M.T.F.

Supporting Online Material

www.sciencemag.org/cgi/content/full/315/5811/518/DC1
Materials and Methods

Fig. S1

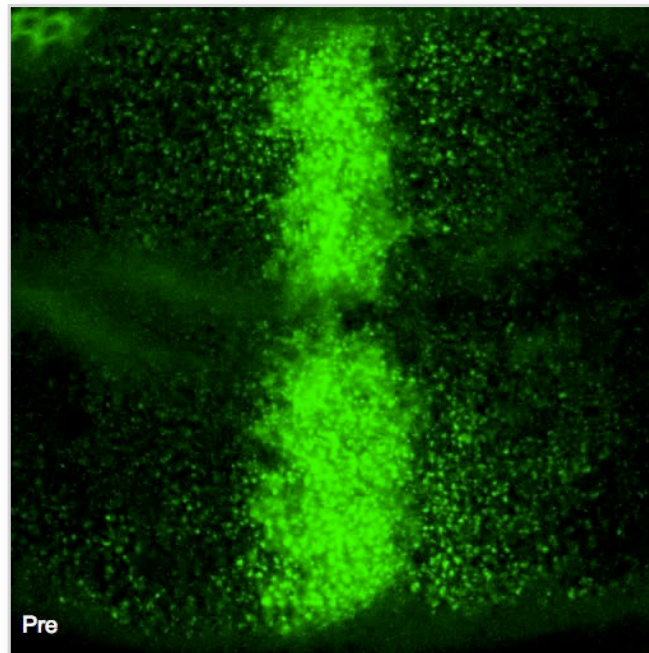
References

8 September 2006; accepted 22 November 2006
10.1126/science.1134910

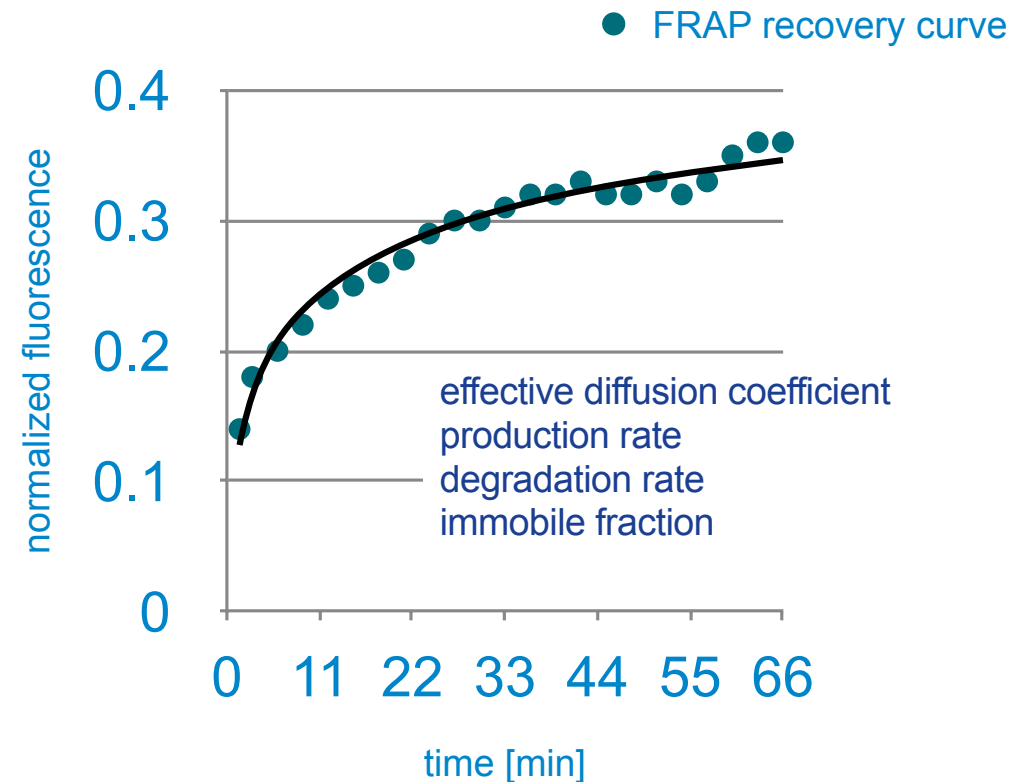
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Fluorescence recovery after photobleaching (FRAP): kinetics of morphogen gradient formation

FRAP of GFP-Dpp

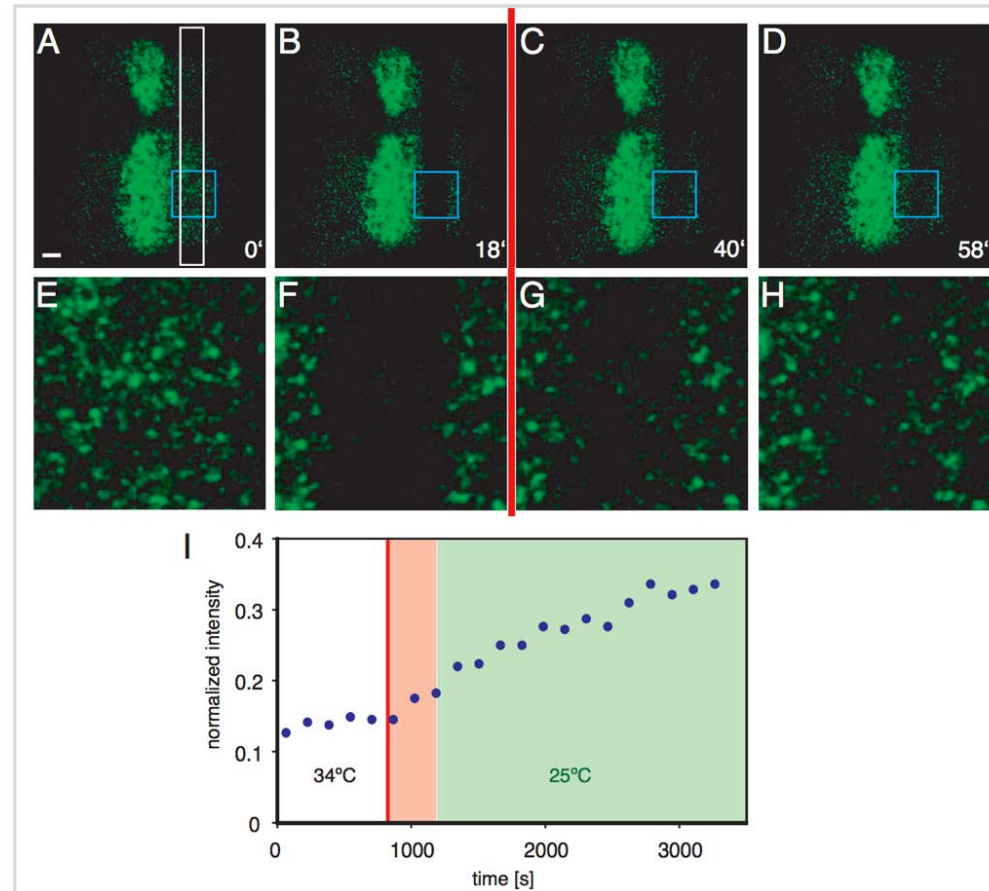


dppGal4::UAS-GFP-Dpp/+



Science. 2007 Jan 26;315(5811):521-5.

The block of endocytosis in the “shibire rescue” experiments is reversible



What mechanism creates the Dpp morphogen gradient of the Drosophila wing disc

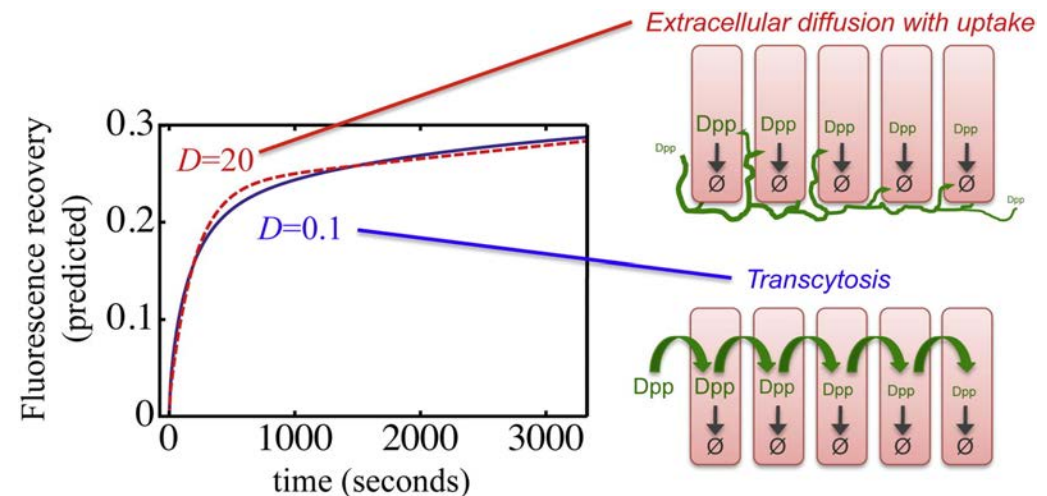
How does the Dpp Morphogen gradient form?

→ Transcytosis?

- FRAP experiments by Kicheva et. al and Gregor et. al.
- Low Diffusion coeff. ($<1 \mu\text{m}^2/\text{s}$)
- Large Dpp fraction diffusing

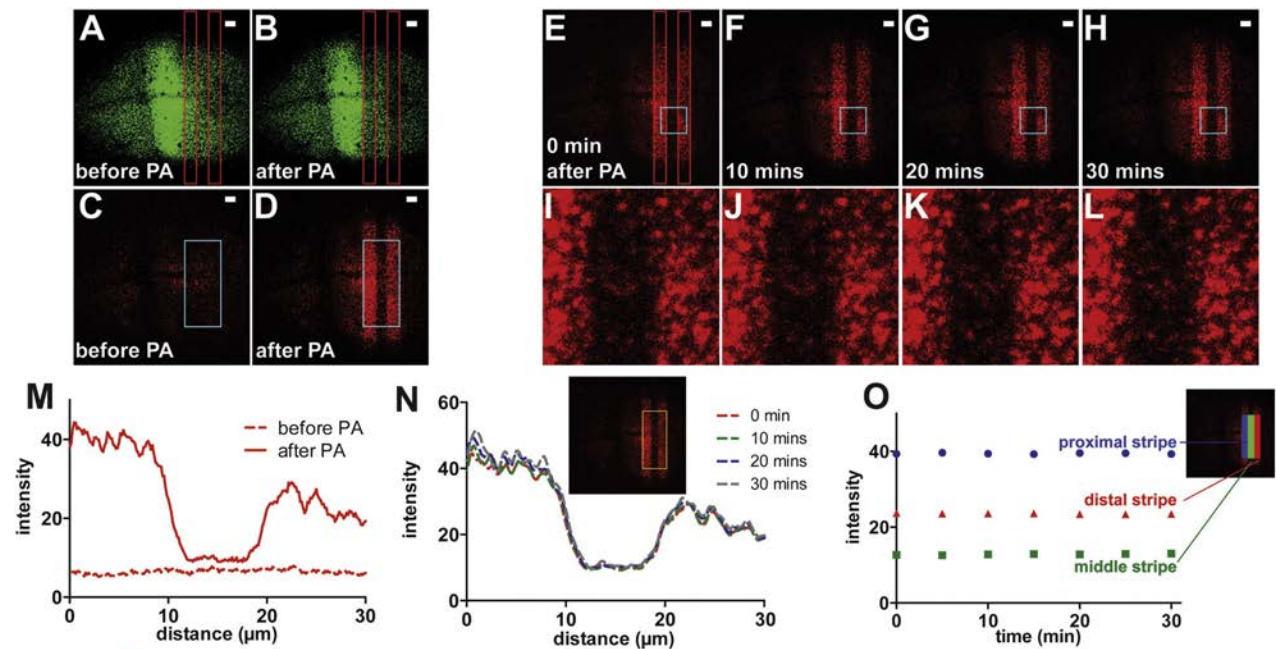
→ Extracellular diffusion?

- Fast, unhindered diffusion
- High Diffusion coeff. ($>20 \mu\text{m}^2/\text{s}$)
- High receptor-mediated uptake and degradation



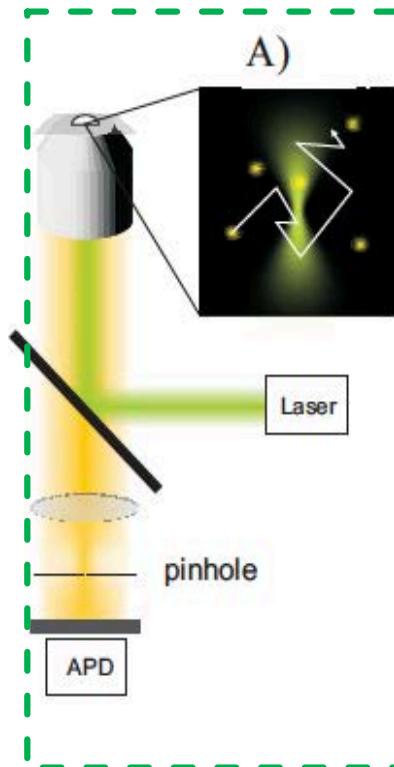
What mechanism creates the Dpp morphogen gradient of the *Drosophila* wing disc

- Dendra2 -> green to red switch
- No significant spreading of red fluorescence

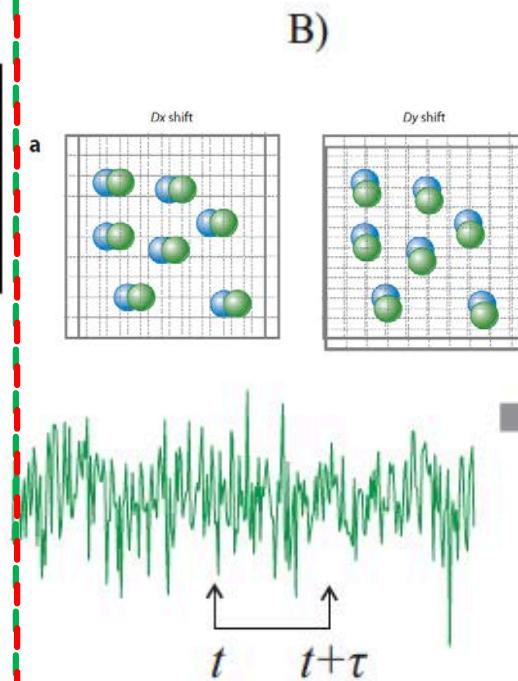


General principle of FCS

1. Confocal microscopy

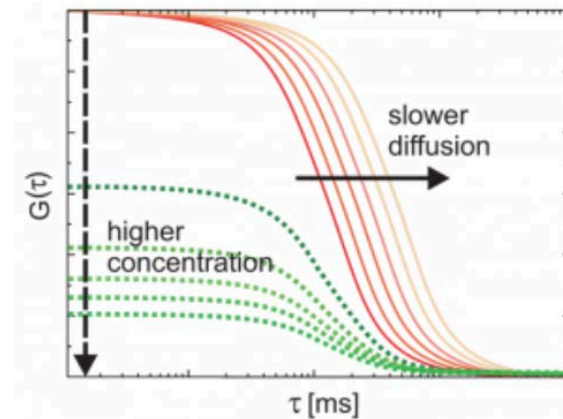


2. Concentration & intensity fluctuation

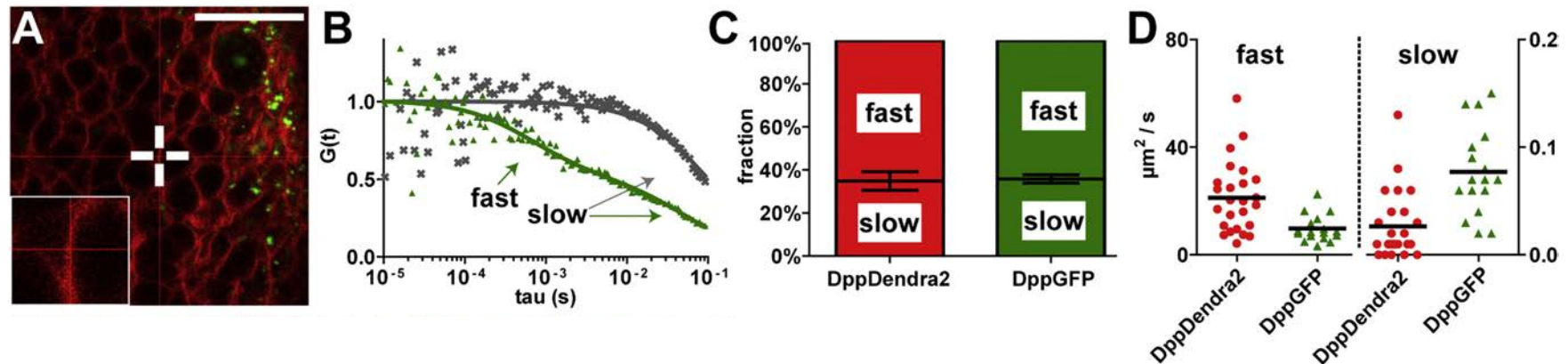


3. Time-correlation analysis

$$G(\tau) = \frac{\langle \delta F(t) \cdot \delta F(t + \tau) \rangle}{\langle F \rangle^2}$$



Two species of Dpp kinetics



- Two species:
- Free moving DppDendra2 (65%)
 - $D = 21 \mu\text{m}^2/\text{s}$
- Membrane bound DppDendra2 (35%)
 - $D \ll 1 \mu\text{m}^2/\text{s}$

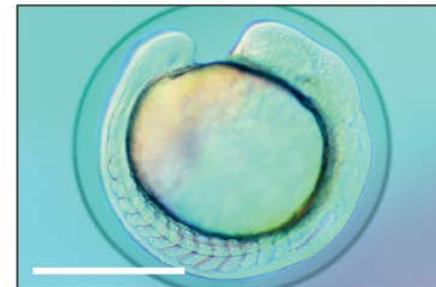
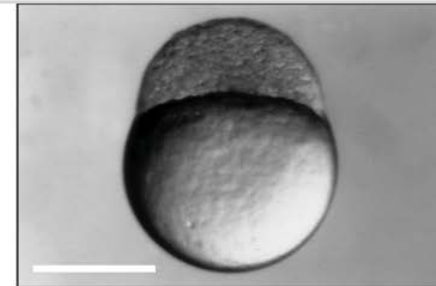
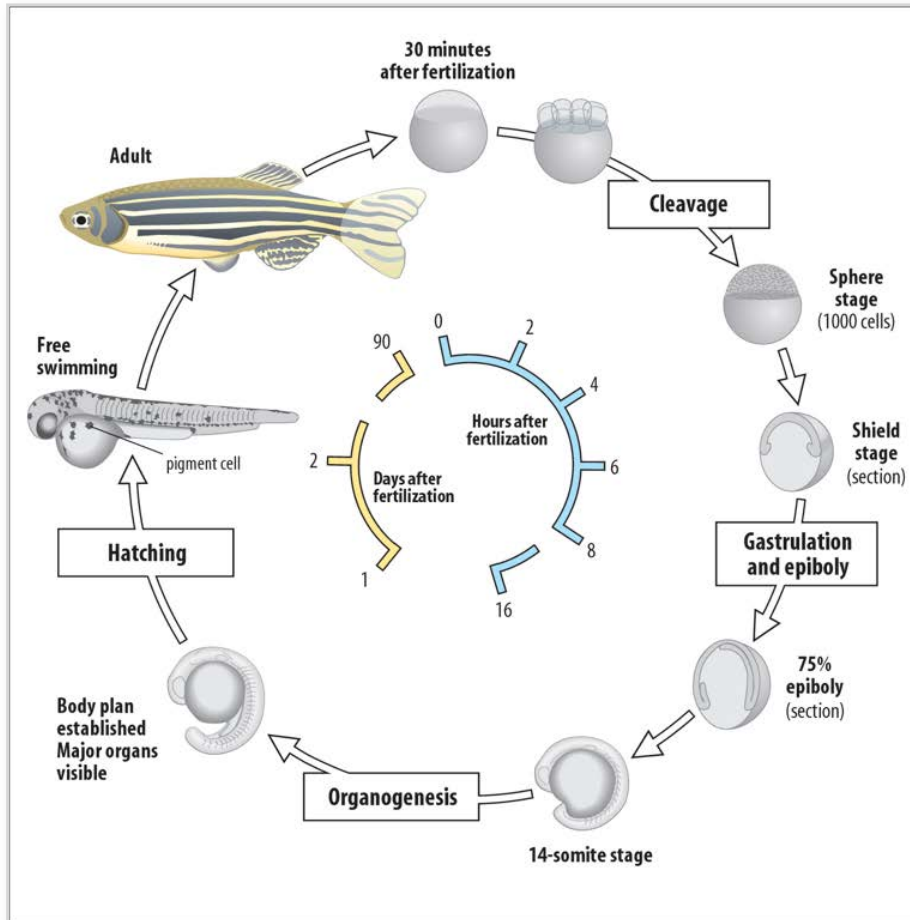
What we will cover

- Quantitative imaging of morphogen dynamics in zebrafish (*Danio rerio*)
 - Reaction-diffusion (Turing) model of pattern generation
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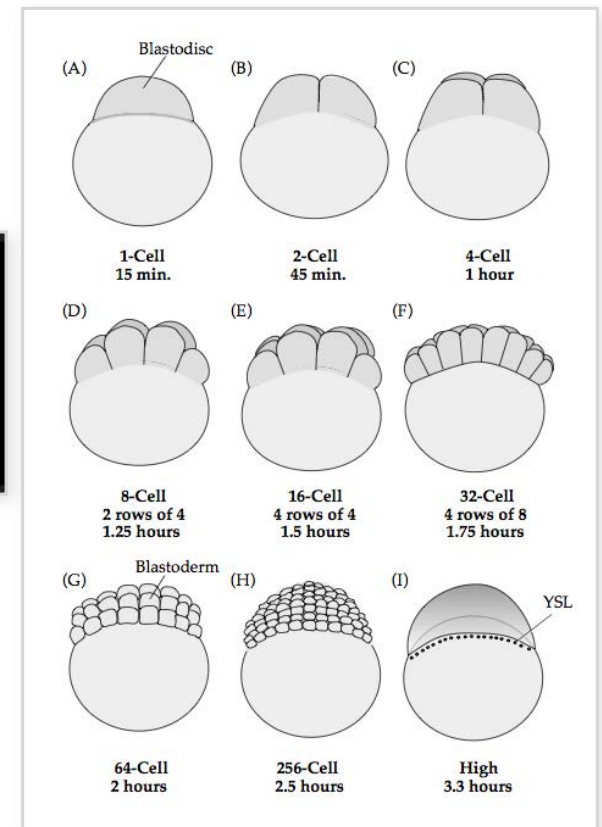
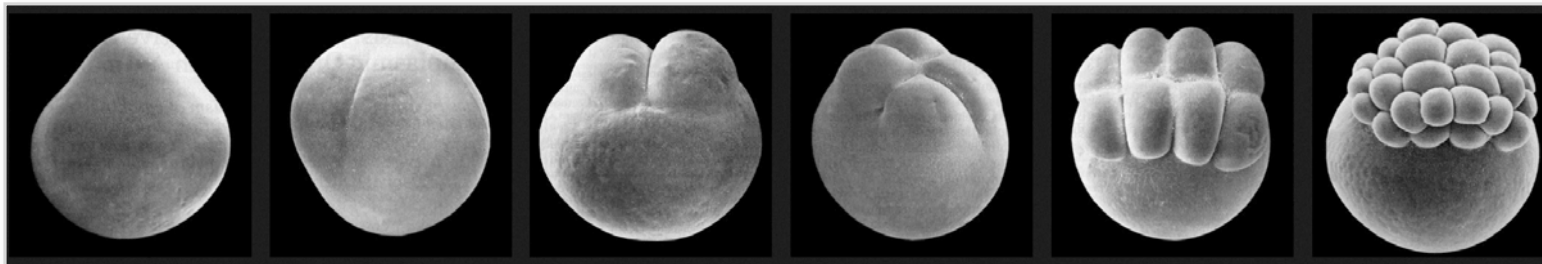
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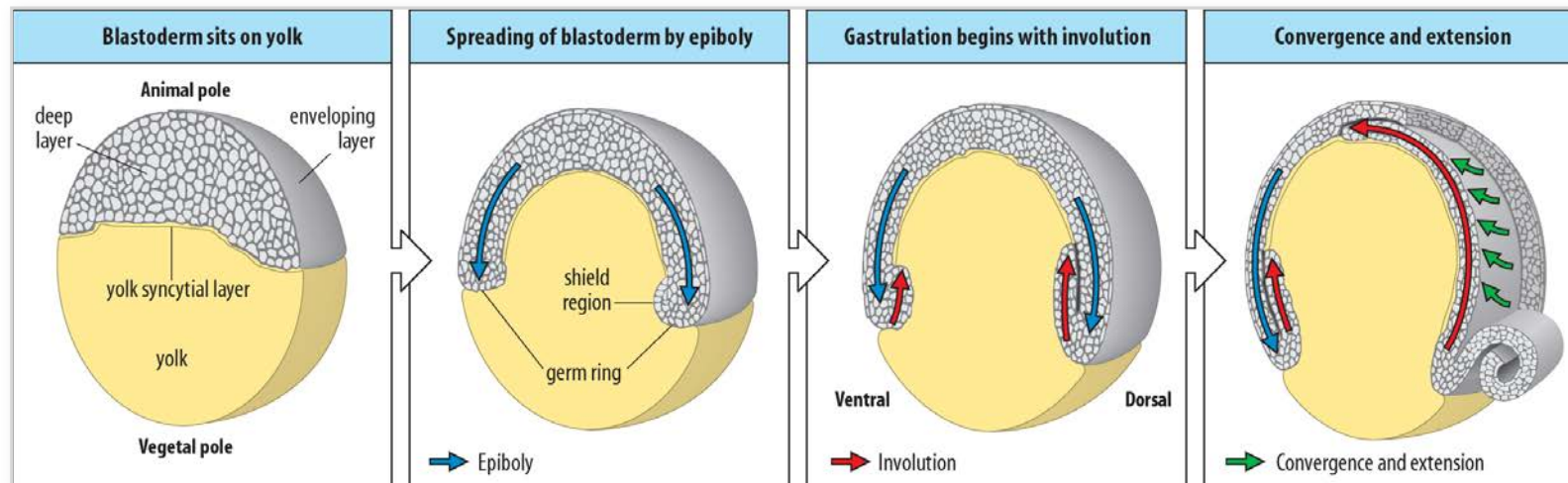
Zebrafish development



Zebrafish development: cleavage occurs at the animal pole



Zebrafish development: epiboly and gastrulation



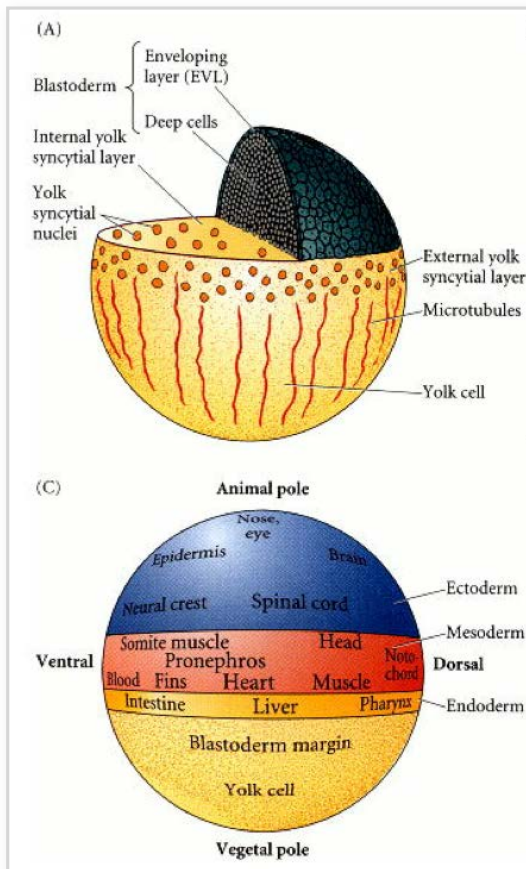
Zebrafish development: all together



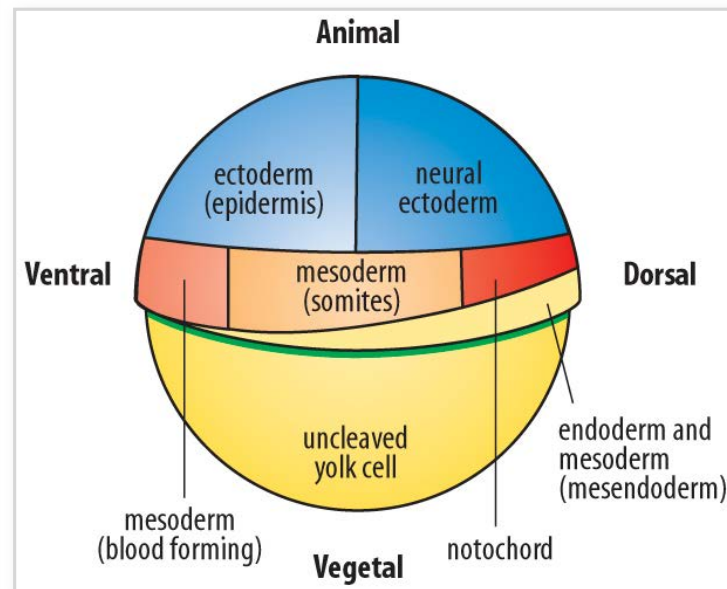
Danio Rerio
(Zebrafish)

Zebrafish development: fate maps

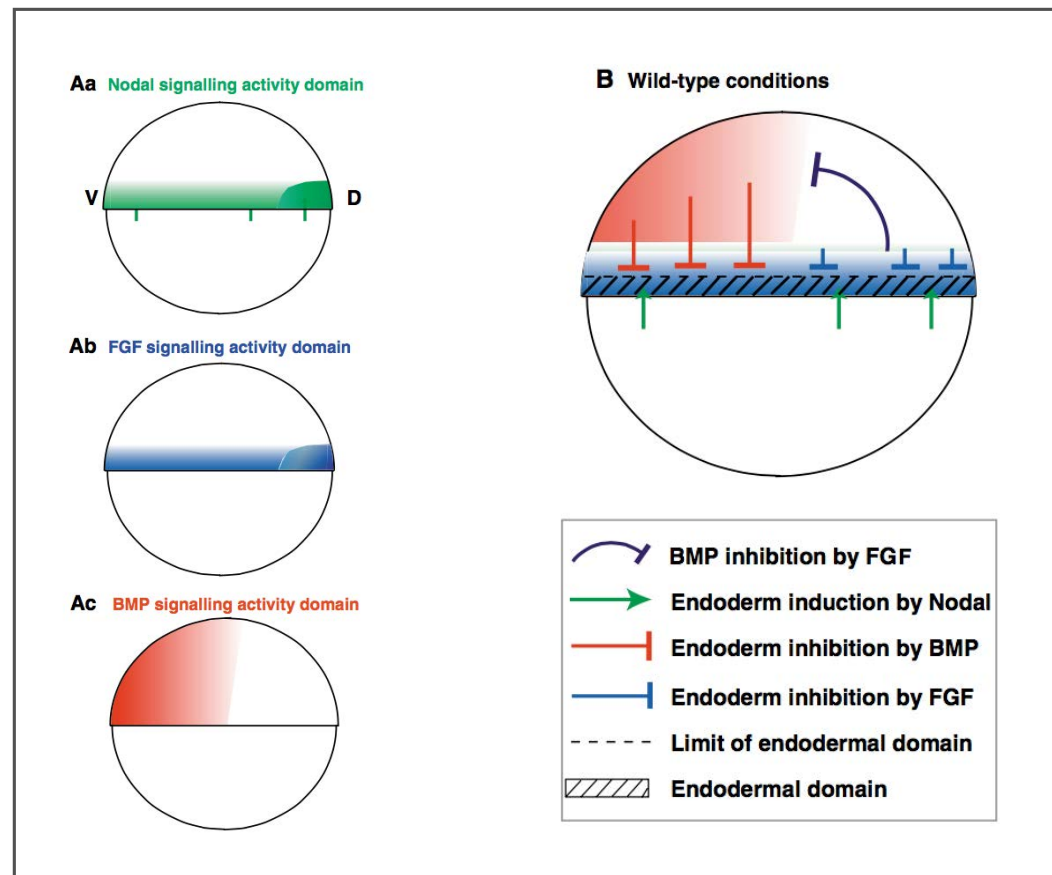
prior to gastrulation



at early gastrula stage



Zebrafish development: fate maps



LETTERS

Fgf8 morphogen gradient forms by a source-sink mechanism with freely diffusing molecules

Shuizi Rachel Yu^{1,3*}, Markus Burkhardt^{2,3*}, Matthias Nowak^{1,3}, Jonas Ries^{2,3}, Zdeněk Petrášek^{2,3}, Steffen Scholpp^{1†}, Petra Schwillé^{2,3} & Michael Brand^{1,3}

It is widely accepted that tissue differentiation and morphogenesis in multicellular organisms are regulated by tightly controlled concentration gradients of morphogens^{1,2}. How exactly these gradients are formed, however, remains unclear^{3–12}. Here we show that Fgf8 morphogen gradients in living zebrafish embryos are established and maintained by two essential factors: fast, free diffusion of single molecules away from the source through extracellular space, and a sink function of the receiving cells, regulated by receptor-mediated endocytosis. Evidence is provided by directly examining single molecules of Fgf8 in living tissue by fluorescence correlation spectroscopy, quantifying their local mobility and concentration with high precision. By changing the degree of uptake of Fgf8 into its target cells, we are able to alter the shape of the Fgf8 gradient. Our results demonstrate that a freely diffusing morphogen can set up concentration gradients in a complex multicellular tissue by a simple source-sink mechanism.

Morphogens are signalling molecules that can induce distinct cellular responses in a concentration-dependent manner^{1,2}. Many examples have previously been described, including fibroblast growth factor8 (Fgf8) in zebrafish embryonic development^{2,13–18}. How morphogens such as Fgf8 move through a tissue and set up concentration gradients remains debated. In the 1970s, Francis Crick proposed that simple diffusion and spatially uniform degradation is sufficient for setting up gradients¹. This mechanism has been considered in various systems^{8–11} and it has been shown that a random walk model can accurately describe the DPP morphogen gradient in *Drosophila* wing discs¹⁹. Moreover, studies have suggested that morphogens can be sequestered from target tissue by receptor-mediated removal^{21,19}. However, a full biophysical description of morphogen gradient formation is still lacking. Meanwhile, alternative models including receptor-aided bucket-brigade mechanism⁴ and directed transport¹¹ have been proposed for morphogen spreading. Here we report a fluorescence correlation spectroscopy (FCS)-based assay for examining single-molecule dynamics of Fgf8 in living zebrafish embryos (Fig. 1a–c and Supplementary Information)^{20,21}. By directly measuring its mobility and concentration gradient, we determine the mechanism of Fgf8 morphogen gradient formation (Supplementary Fig. 1).

We first tagged Fgf8 with enhanced green fluorescent protein (Fgf8-EGFP; Fig. 2a). At equimolar concentrations, the fusion protein showed a similar capacity to induce target gene expression as untagged Fgf8 (Supplementary Fig. 2a, b). Fgf8-EGFP was readily taken up by target cells, and was recruited onto cell membranes by Fgf receptors (Fig. 2a–c)¹. In contrast, a negative control of secreted EGFP (secEGFP) was not. Furthermore, we verified by western blot that the full-length fusion protein was the only EGFP-containing species in the extracellular space of *fgf8-egfp* messenger-RNA-injected embryos (Supplementary Fig. 2c).

To investigate how Fgf8 spreads into its target tissue, we took FCS measurements in the extracellular space of gastrulating zebrafish embryos that had a restricted source of the fluorescent protein (Figs 1 and 2b, c and Methods Summary). The FCS autocorrelation curves fit well to a three-dimensional diffusion model, but not with alternative models such as directional active transport only (Fig. 3a and Supplementary Information). Most of the molecules (91%) had a diffusion coefficient ($53 \mu\text{m}^2 \text{s}^{-1}$) that was on the same order of magnitude as that of secEGFP ($86 \mu\text{m}^2 \text{s}^{-1}$; Fig. 3a, b and Supplementary Fig. 3a, b). Fgf8-EGFP moved slightly slower than secEGFP, as is expected for a larger molecule of about double the molecular mass. For a receptor-mediated bucket-brigade mechanism⁴, we would expect most of the ligand to have a diffusion coefficient similar to that of the lateral mobility of the Fgf receptor in the cell membrane, which is about 100-fold smaller²² than the one we measured for Fgf8, thus arguing against this mechanism for Fgf8 spreading. The Fgf8-EGFP diffusion coefficient in the zebrafish embryos is two orders of magnitude larger than that reported for DPP ($0.10 \mu\text{m}^2 \text{s}^{-1}$) in *Drosophila* wing discs¹⁹. This supports the notion that the two morphogens have very different propagation mechanisms in their respective target tissues. DPP is transported by transcytosis involving repeated cycles of endocytosis and exocytosis through target cells^{5,10}. This is likely to occur much slower than the free diffusion of Fgf8-EGFP in the extracellular space.

In addition, we analysed recombinant Fgf8 protein labelled with the organic dye Cy5. The autocorrelation curve of Fgf8-Cy5 fitted

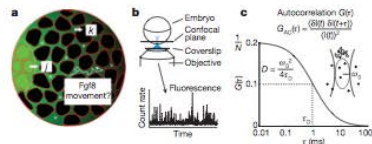
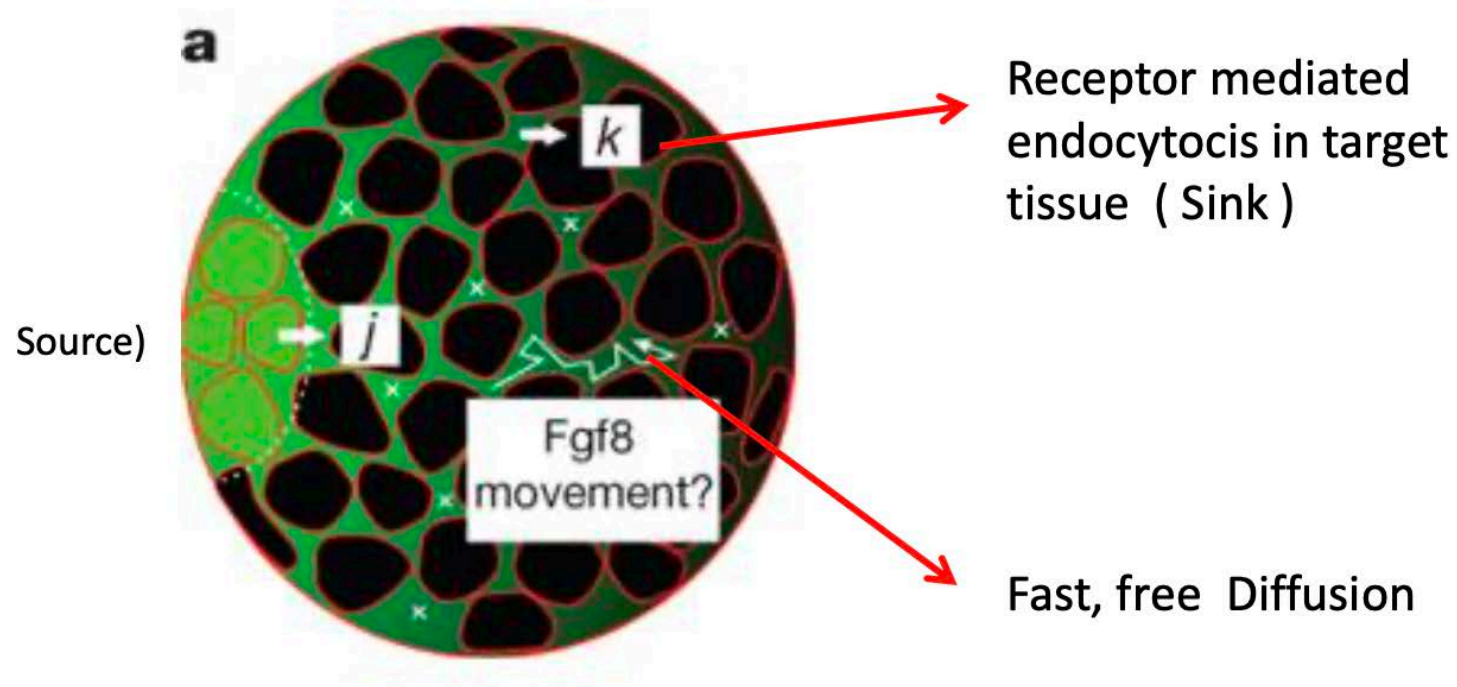


Figure 1 | FCS setup to investigate morphogen characteristics. **a**, Schematic section of a gastrulating zebrafish embryo. Morphogen is produced from a restricted source (green cells) with current j , spreads into target tissue (black cells) and is removed from the extracellular space with degradation rate k . Crosses indicate extracellular spaces where FCS measurements are taken. **b**, One-focus FCS setup. **c**, The autocorrelation (AC) function $G(\tau)$ has two key parameters: (1) the characteristic dwell time τ_D , which relates to the diffusion coefficient D ; and (2) the amplitude $G(0)$, which is inversely proportional to the number of fluorescent particles N in the detection volume (Supplementary Information).

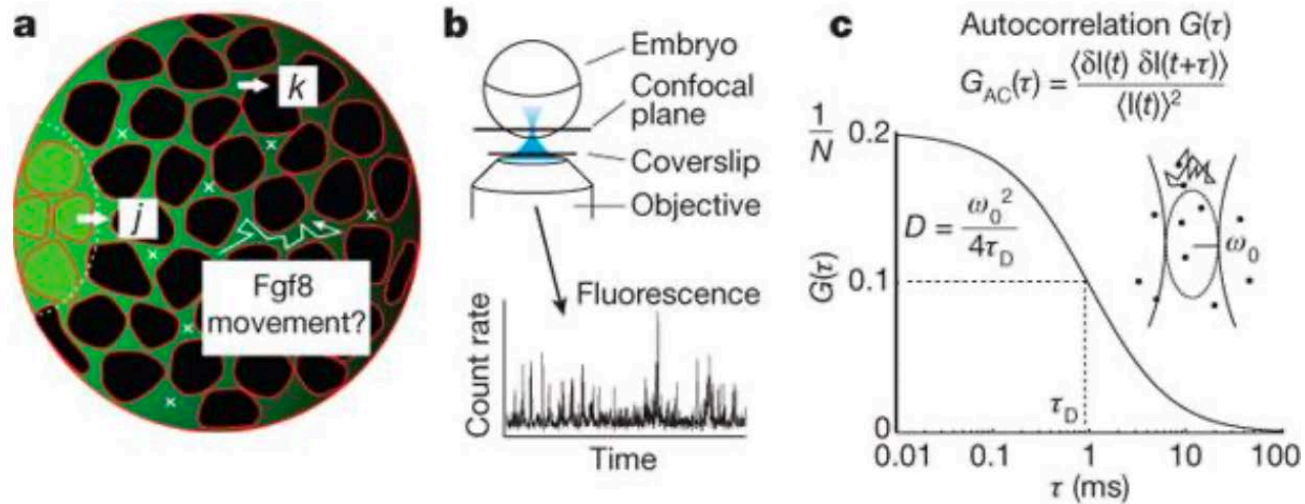
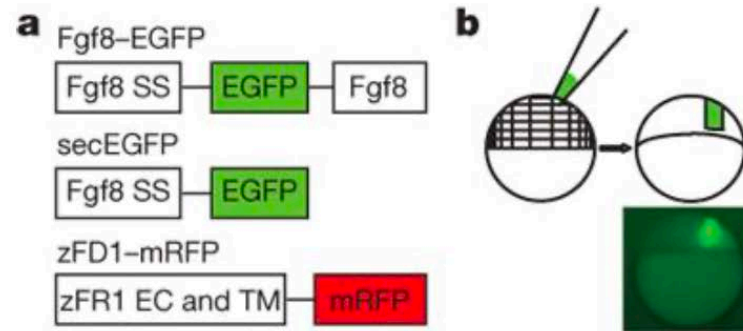
¹Developmental Genetics, Biotechnology Center, TUD, ²Biophysics, Biotechnology Center, TUD, and ³Center for Regenerative Therapies, TUD, Tatzberg 47-49, 01307 Dresden, Germany. [†]Present address: Institute of Toxicology and Genetics, Forschungszentrum Karlsruhe, D-76021 Karlsruhe, Germany.

*These authors contributed equally to this work.

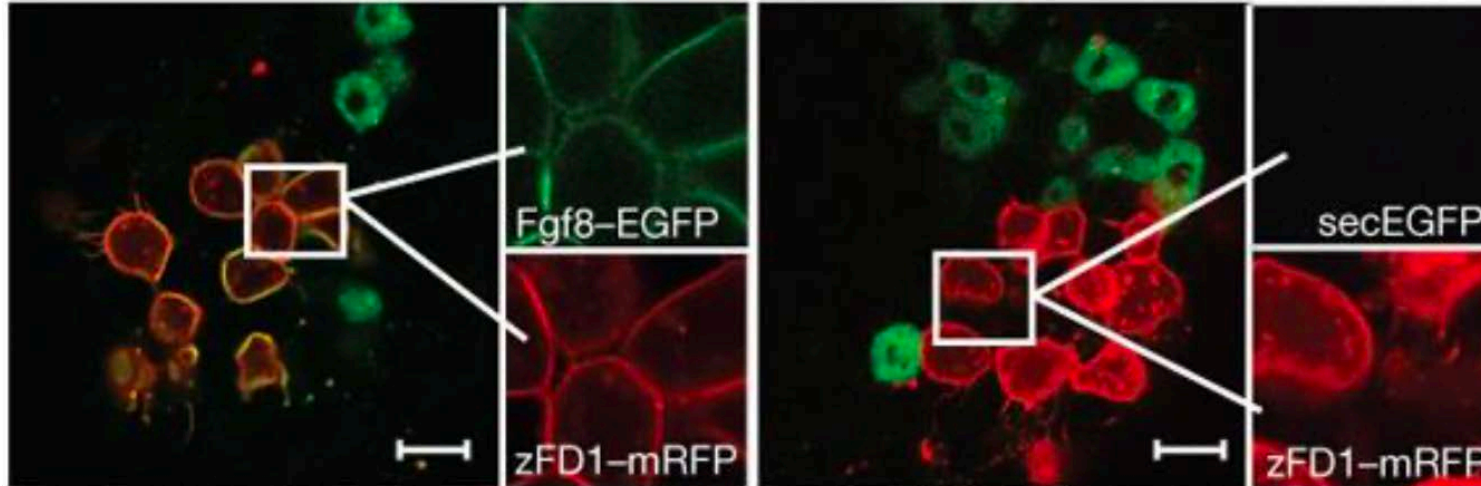
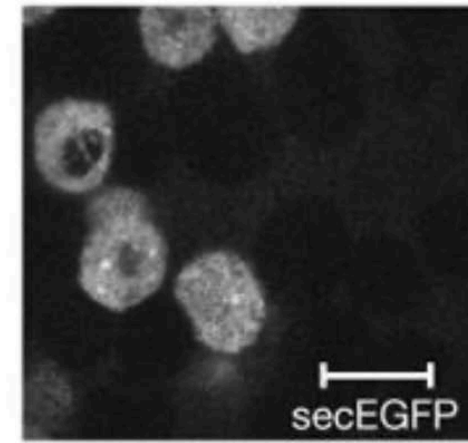
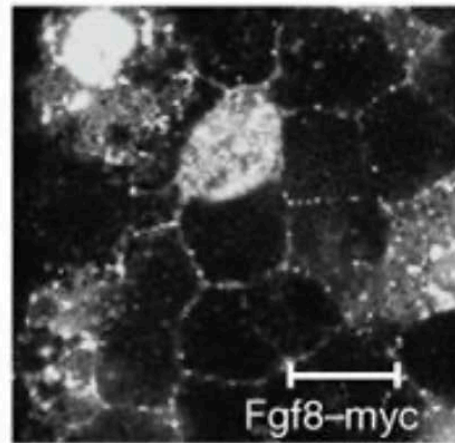
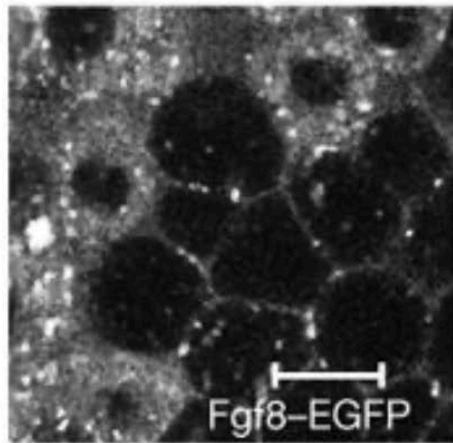
Source-Sink Mechanism?



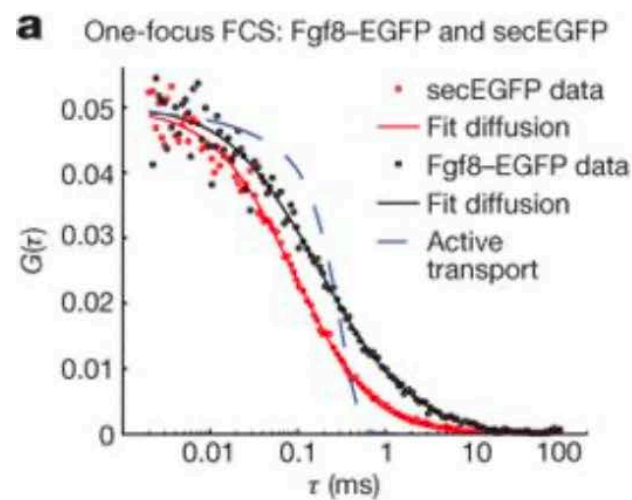
Experimental setup



Various fusion proteins



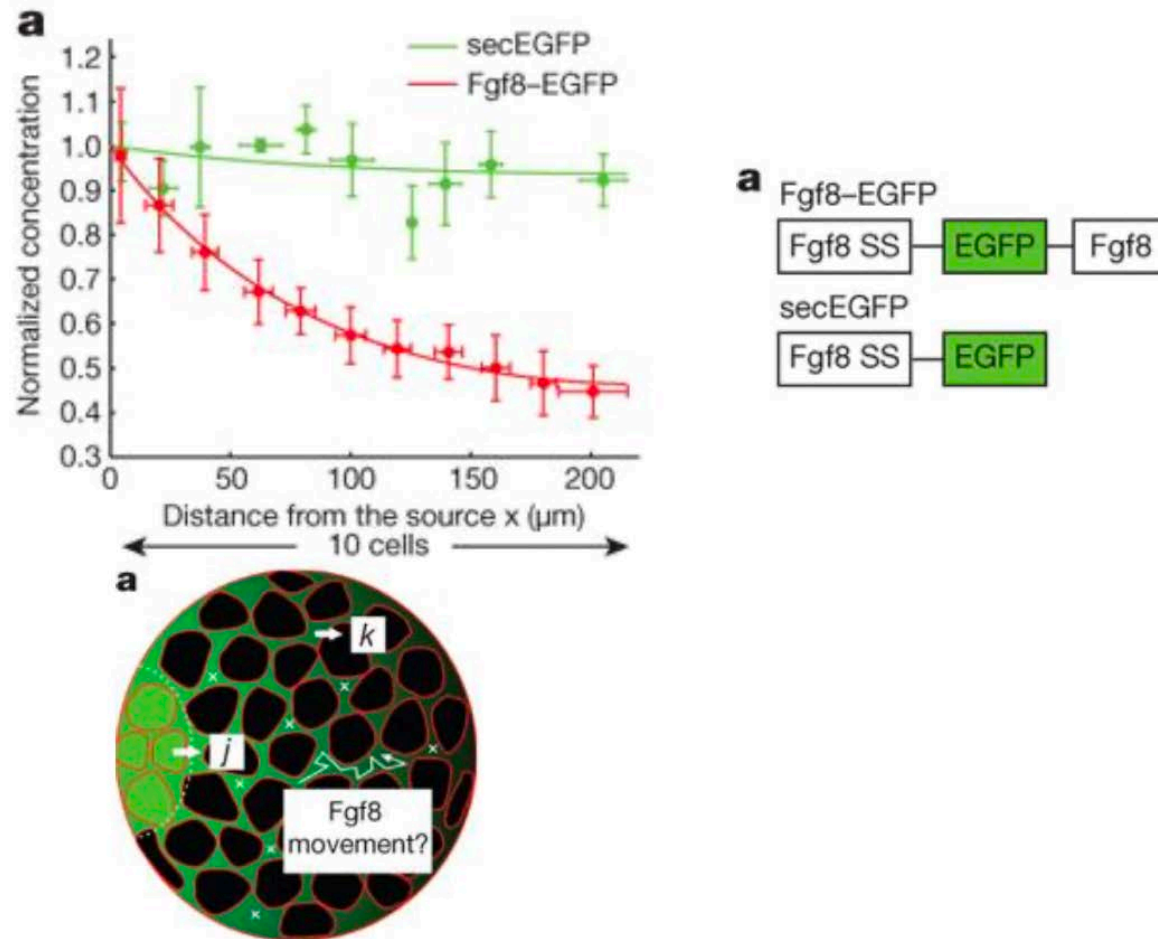
Free 3D Diffusion dominates the movement



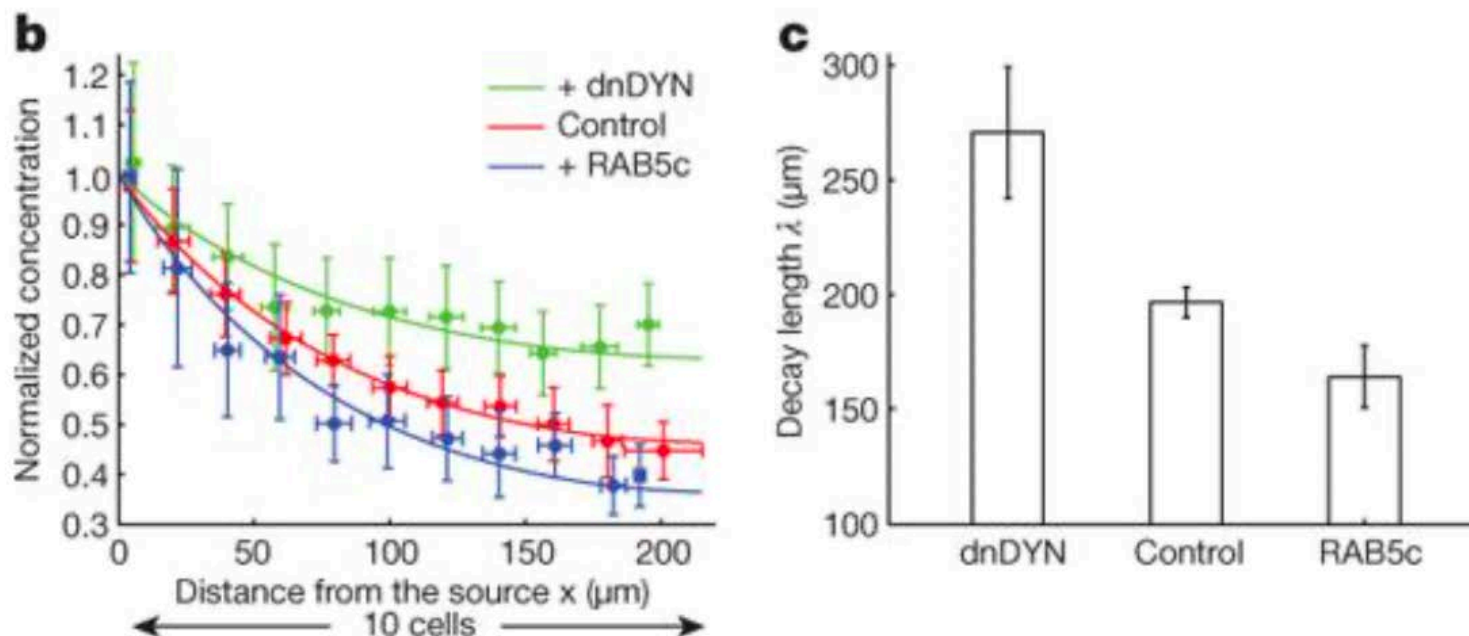
b

	F (%)	Diffusion coefficient ($\mu\text{m}^2 \text{s}^{-1}$)		MM (kDa)	n
		Fast component	Slow component		
secEGFP	99 ± 1	86 ± 11	NA	28	29
Fgf8-EGFP	91 ± 5	53 ± 8	4 ± 3	50	29
Cy5	99 ± 2	190 ± 6	NA	1	8
Fgf8-Cy5	80 ± 12	91 ± 26	7 ± 2	23	30

The Fgf8-gradient profile decays exponentially



Modulating the Fgf8-gradient



Endocytosis regulates extracellular FGF8 concentration gradient
Data support a source sink model for gradient formation

What we will cover

- Quantitative imaging of morphogen dynamics in zebrafish (*Danio rerio*)
 - **Reaction-diffusion (Turing) model of pattern generation**
-

Reaction-diffusion (Turing) model of pattern generation

THE CHEMICAL BASIS OF MORPHOGENESIS

By A. M. TURING, F.R.S. *University of Manchester*

(Received 9 November 1951—Revised 15 March 1952)

It is suggested that a system of chemical substances, called morphogens, reacting together and diffusing through a tissue, is adequate to account for the main phenomena of morphogenesis. Such a system, although it may originally be quite homogeneous, may later develop a pattern or structure due to an instability of the homogeneous equilibrium, which is triggered off by random disturbances. Such reaction-diffusion systems are considered in some detail in the case of an isolated ring of cells, a mathematically convenient, though biologically unusual system. The investigation is chiefly concerned with the onset of instability. It is found that there are six essentially different forms which this may take. In the most interesting form stationary waves appear on the ring. It is suggested that this might account, for instance, for the tentacle patterns on *Hydra* and for whorled leaves. A system of reactions and diffusion on a sphere is also considered. Such a system appears to account for gastrulation. Another reaction system in two dimensions gives rise to patterns reminiscent of dappling. It is also suggested that stationary waves in two dimensions could account for the phenomena of phyllotaxis.

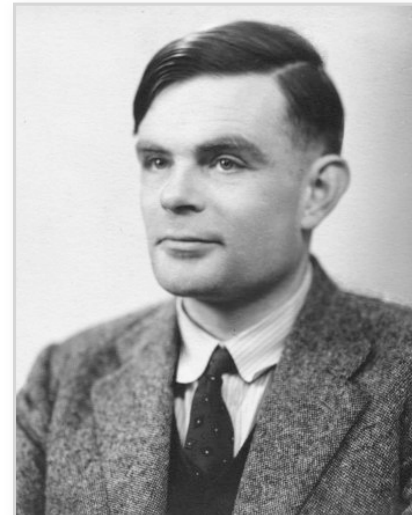
The purpose of this paper is to discuss a possible mechanism by which the genes of a zygote may determine the anatomical structure of the resulting organism. The theory does not make any new hypotheses; it merely suggests that certain well-known physical laws are sufficient to account for many of the facts. The full understanding of the paper requires a good knowledge of mathematics, some biology, and some elementary chemistry. Since readers cannot be expected to be experts in all of these subjects, a number of elementary facts are explained, which can be found in text-books, but whose omission would make the paper difficult reading.

1. A MODEL OF THE EMBRYO. MORPHOGENS

In this section a mathematical model of the growing embryo will be described. This model will be a simplification and an idealization, and consequently a falsification. It is to be hoped that the features retained for discussion are those of greatest importance in the present state of knowledge.

The model takes two slightly different forms. In one of them the cell theory is recognized but the cells are idealized into geometrical points. In the other the matter of the organism is imagined as continuously distributed. The cells are not, however, completely ignored, for various physical and physico-chemical characteristics of the matter as a whole are assumed to have values appropriate to the cellular matter.

With either of the models one proceeds as with a physical theory and defines an entity called 'the state of the system'. One then describes how that state is to be determined from the state at a moment very shortly before. With either model the description of the state consists of two parts, the mechanical and the chemical. The mechanical part of the state describes the positions, masses, velocities and elastic properties of the cells, and the forces between them. In the continuous form of the theory essentially the same information is given in the form of the stress, velocity, density and elasticity of the matter. The chemical part of the state is given (in the cell form of theory) as the chemical composition of each



Alan Turing
(1912-1954)

Reaction-diffusion (Turing) model of pattern generation

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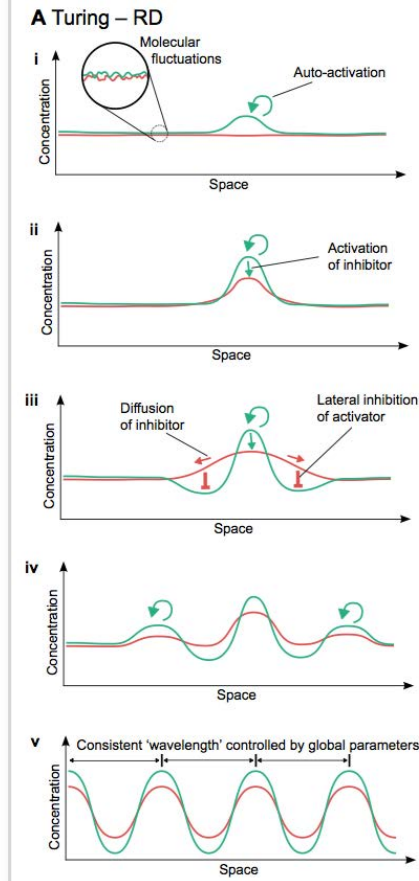
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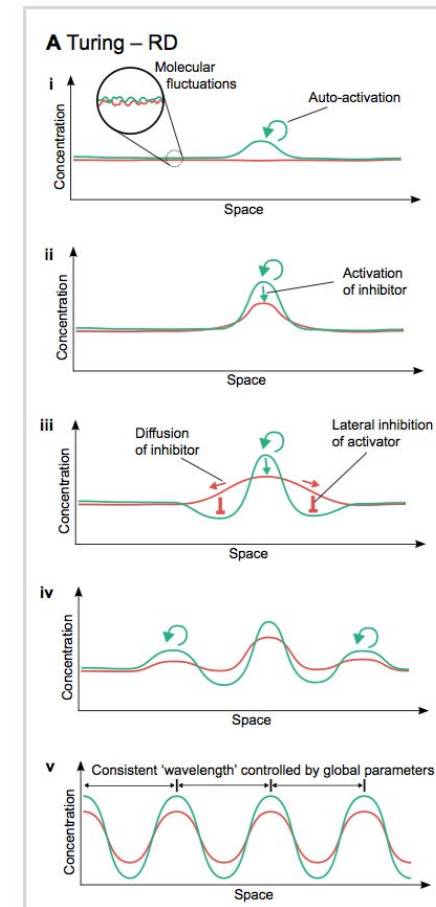
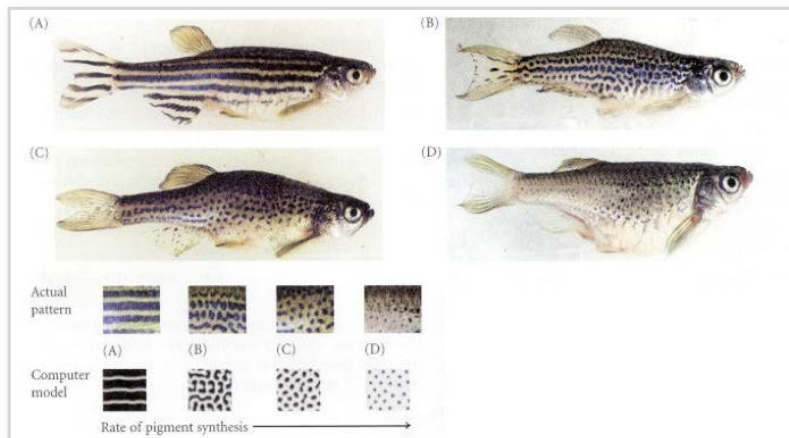
In this section a mathematical model of the growing embryo will be described. This model will be a simplification and an idealization, and consequently a falsification. It is to be hoped that the features retained for discussion are those of greatest importance in the present state of knowledge.

The model takes two slightly different forms. In one of them the cell theory is recognized but the cells are idealized into geometrical points. In the other the matter of the organism is imagined as continuously distributed. The cells are not, however, completely ignored, for various physical and physico-chemical characteristics of the matter as a whole are assumed to have values appropriate to the cellular matter.

With either of the models one proceeds as with a physical theory and defines an entity called 'the state of the system'. One then describes how that state is to be determined from the state at a moment very shortly before. With either model the description of the state consists of two parts, the mechanical and the chemical. The mechanical part of the state describes the positions, masses, velocities and elastic properties of the cells, and the forces between them. In the continuous form of the theory essentially the same information is given in the form of the stress, velocity, density and elasticity of the matter. The chemical part of the state is given (in the cell form of theory) as the chemical composition of each



Reaction-diffusion (Turing) model of pattern generation



undetectable in chondrocytes; CBF β is expressed in all cell types at comparable levels (figs. S13 and S14). Fluorescent immunostaining of mouse joint tissue sections also showed coexpression of CBF β and RUNX1 in cartilage cells (fig. S15). These results are consistent with previous findings about the roles of the genes in chondrogenesis and osteogenesis, and lead us to speculate that RUNX1, rather than RUNX2 or RUNX3, is probably the downstream effector of KGN. When we knocked down RUNX1 expression in hMSCs with shRNAs, KGN-induced chondrocyte differentiation was blocked (fig. S16). Furthermore, *RUNX2* transcription has been shown to be suppressed by RUNX2 through an autoregulation mechanism in different types of cells (37). Given the fact that all RUNX family members recognize the same DNA sequence, it is plausible that the RUNX1-CBF β complex also suppresses *RUNX2* transcription, thus keeping RUNX2 at a relatively low level, which inhibits osteoblast and terminal chondrocyte differentiation.

The development of disease-modifying osteoarthritis drugs has to date focused on (i) slowing the erosion of cartilage by inhibiting matrix-degrading enzymes; (ii) protecting chondrocytes from catabolic activity with antagonists of interleukin-1 β and TNF- α or inhibitors of p38, MEK, and caspases; and (iii) the development of anabolic agents, such as FGF18, which promote chondrocyte and chondrocyte progenitor cell proliferation (3, 32, 33). This work is based on small-molecule effectors of endogenous stem cell populations and may prove synergistic with other

efforts currently in development. KGN functions by binding FLNA and disrupting its interaction with CBF β , which in turn modulates the RUNX family of transcription factors. Recently, two other examples have been reported (34, 35) in which small druglike molecules selectively regulate transcription factor subcellular localization and downstream transcriptional activities. Thus, small molecules appear to be useful tools for selectively modulating gene transcription in cells and whole organisms, and in this case have provided new insights into chondrocyte biology that may lead to novel therapies for the treatment of this degenerative disease.

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Supplementary Materials

www.sciencemag.org/cgi/content/full/science.1215157/DC1

Materials and Methods

Figs. S1 to S16

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Differential Diffusivity of Nodal and Lefty Underlies a Reaction-Diffusion Patterning System

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Biological systems involving short-range activators and long-range inhibitors can generate complex patterns. Reaction-diffusion models postulate that differences in signaling range are caused by differential diffusivity of inhibitor and activator. Other models suggest that differential clearance underlies different signaling ranges. To test these models, we measured the biophysical properties of the Nodal/Lefty activator/inhibitor system during zebrafish embryogenesis. Analysis of Nodal and Lefty gradients revealed that Nodals have a shorter range than Lefty proteins. Pulse-labeling analysis indicated that Nodals and Leftys have similar clearance kinetics, whereas fluorescence recovery assays revealed that Leftys have a higher effective diffusion coefficient than Nodals. These results indicate that differential diffusivity is the major determinant of the differences in Nodal/Lefty range and provide biophysical support for reaction-diffusion models of activator/inhibitor-mediated patterning.

In 1952, Alan Turing put forward the reaction-diffusion model, in which two interacting and diffusing species of molecules can generate complex patterns (1). Gierer and Meinhardt postulated that pattern formation in reaction-diffusion

models requires a short-range activator that enhances both its own production and that of a long-range inhibitor (2) (Fig. 1A). Despite the prominence of reaction-diffusion models and the widespread occurrence of short-range acti-

vators and long-range inhibitors in development (3–10), it is unclear how differences in activator and inhibitor ranges arise in vivo. The classic reaction-diffusion models postulate that the inhibitor is more diffusive than the activator (text S1), but more recent studies suggest that differential signal clearance might be a major determinant of differences in signaling range (11–18) (Fig. 1B). This question has not been resolved because the biophysical properties of diffusion and clearance have not been measured for any activator/inhibitor pair.

The transforming growth factor- β superfamily signals Nodal and Lefty constitute an activator/inhibitor-based system in animals as different as sea urchin and mouse (3–5, 16, 18–22) (text S2). Nodals activate signaling during mesoderm induction and left-right patterning,

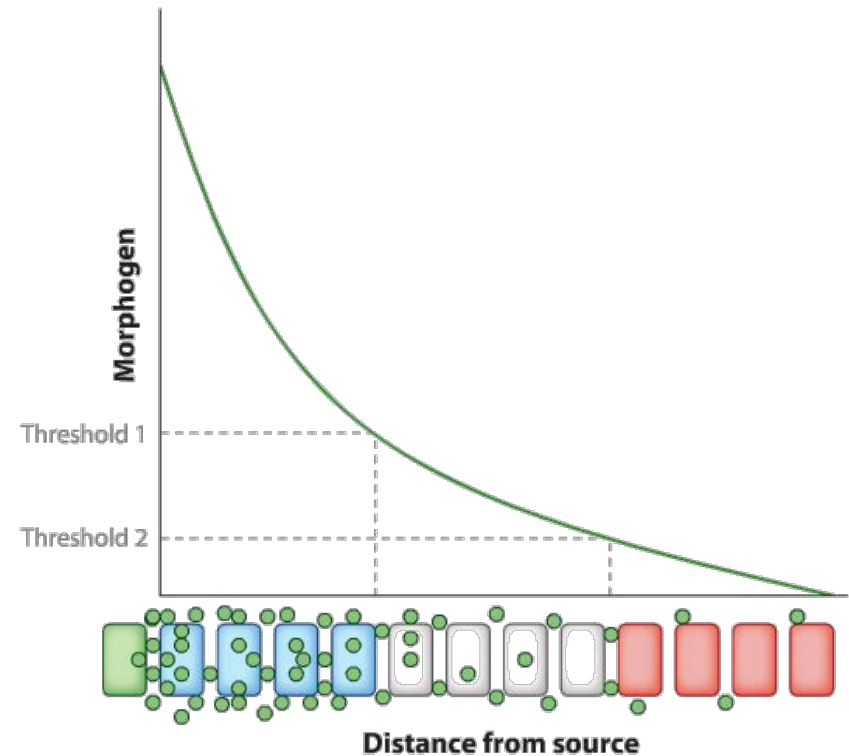
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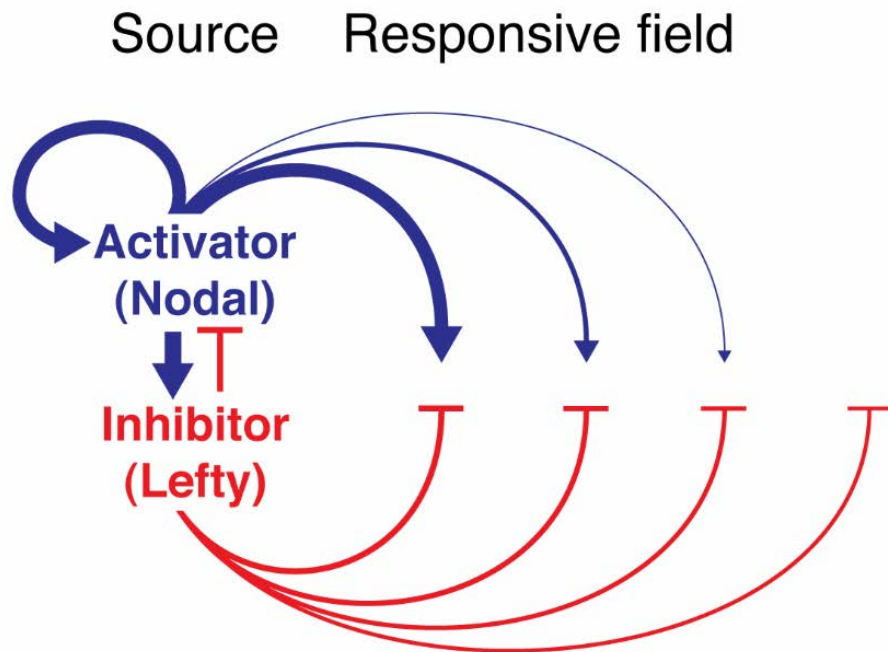
†These authors contributed equally to this work.

Synthesis-Diffusion-Clearance Model

- Produced localised source, diffuses through tissues, and is cleared
- Distance of gradient decay depends on the diffusivity and clearance rates

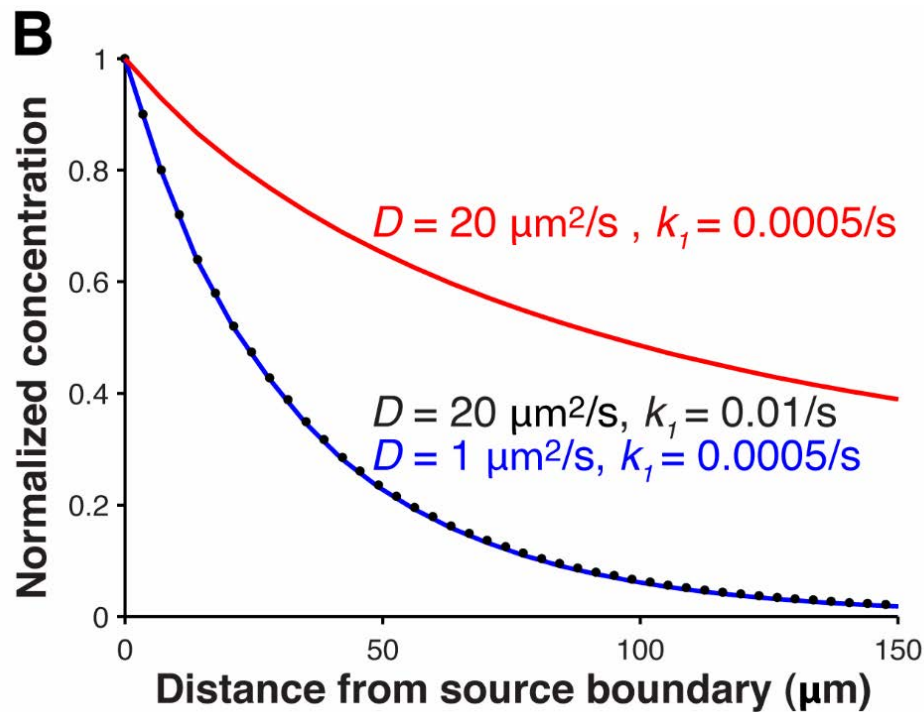


A theory of biological pattern formation



- Gierer and Meinhardt 1972
- Short-range activator enhances both its own production and long-range inhibitor
- In Vivo Mechanism Unclear

Morphogen gradient established by Diffusion or Clearance?



- Classic model assume inhibitor diffusivity major factor
- Biological Studies may show differential clearance

Morphogen gradient established by Diffusion or Clearance?

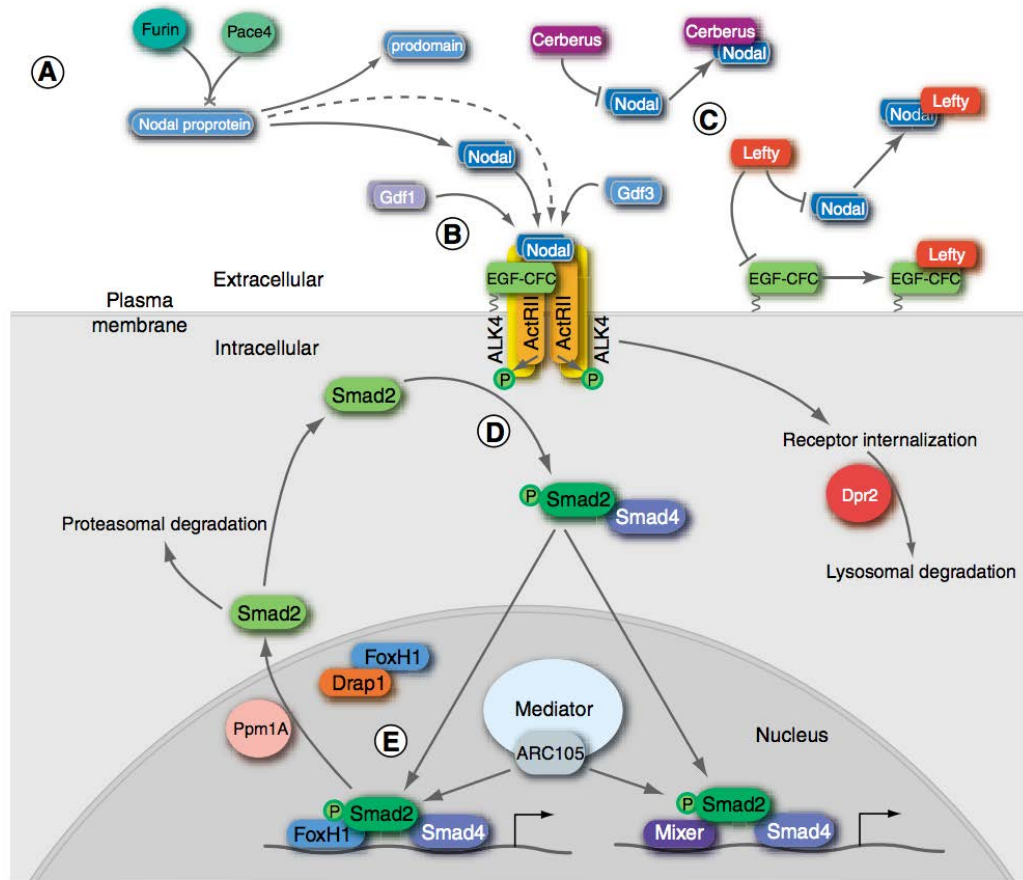
Nodals

Nodals are short- to mid-range activators that enhance their own expression

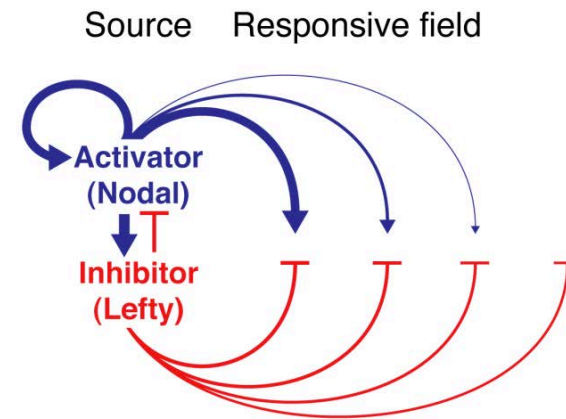
Leftys

Leftys are long-range inhibitors that are activated by Nodals

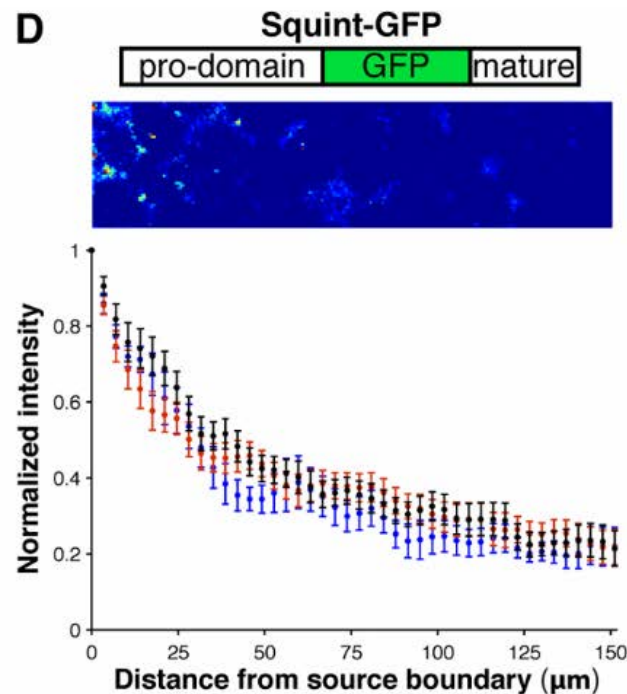
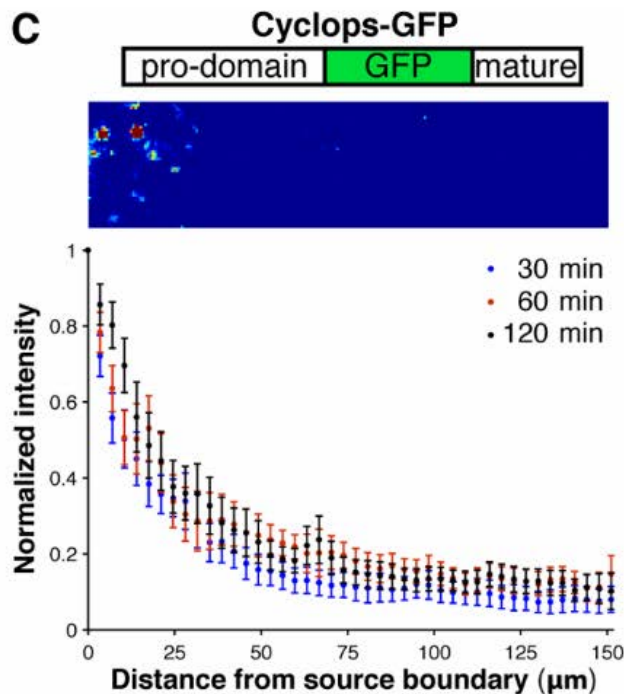
The Nodal signalling pathway



- Model of the **Nodal/Lefty** activator/inhibitor reaction-diffusion system and regulation of range



Visualisation of Nodal and Lefty in vivo: GFP fusion proteins



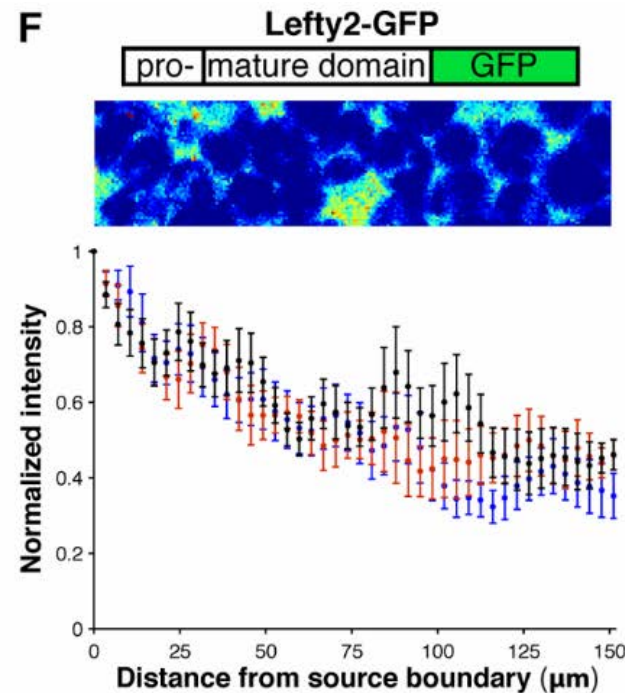
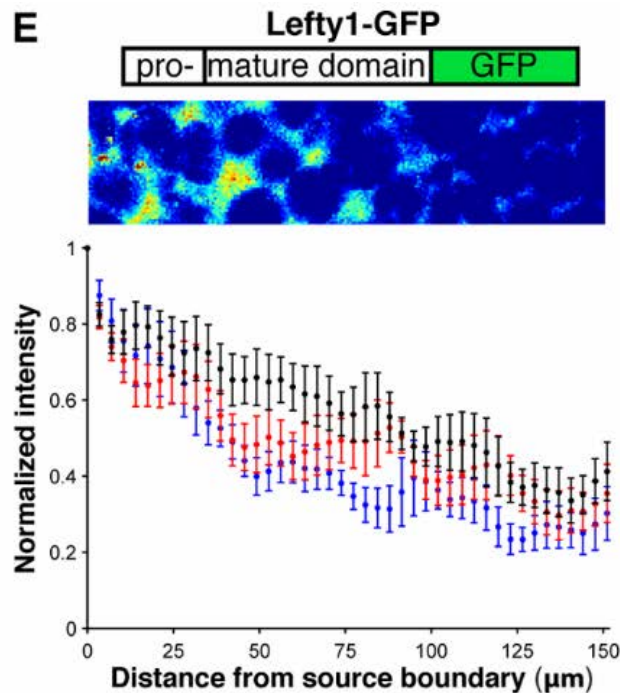
Cyclops-GFP:

- Punctate distribution
- Dropped steeply as the distance from the source increased

Squint-GFP

- more diffuse and reached farther

Visualisation of Nodal and Lefty in vivo: GFP fusion proteins



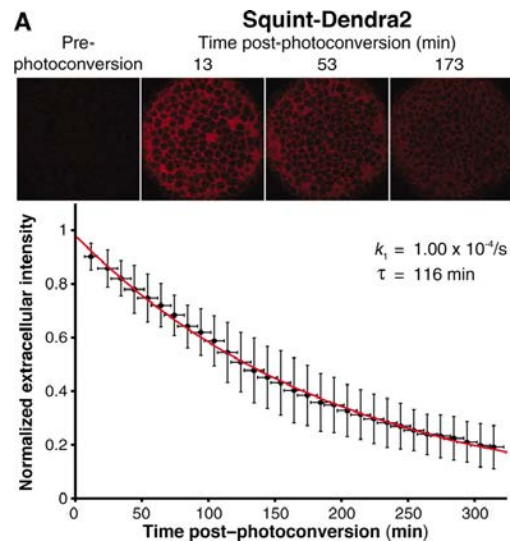
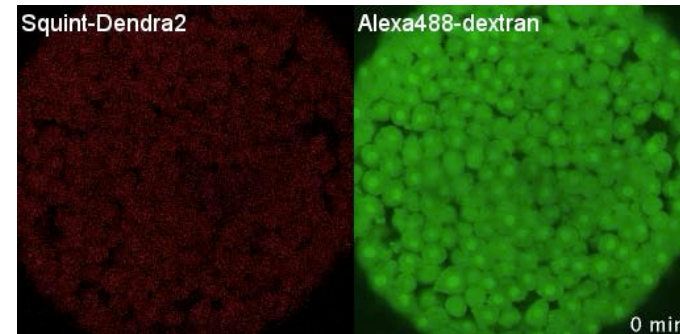
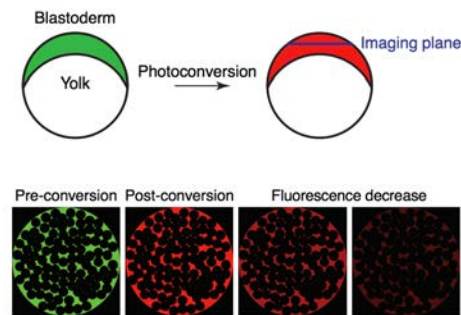
Lefty1-GFP

- Long range of expression
- Flatter distribution

Lefty2-GFP

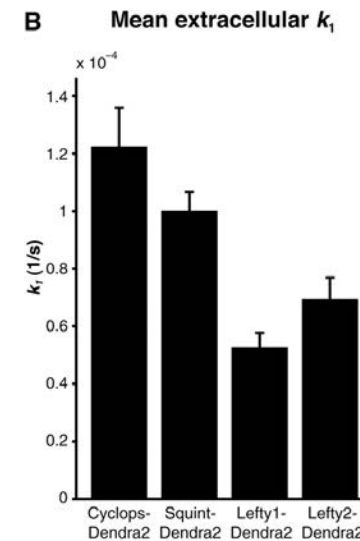
- Super long range of expression

Measuring extracellular clearance rate constants with pulse-labelling assay

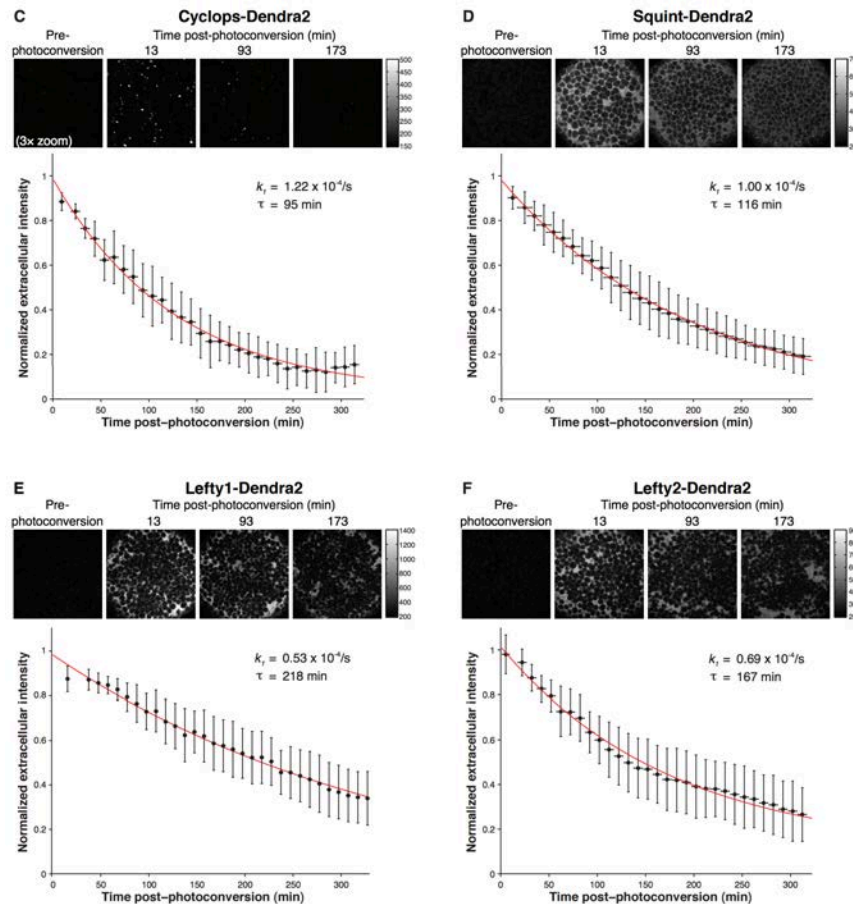


$$\frac{dc}{dt} = -k_1 c$$

$$\bar{\tau} = \frac{\ln(2)}{\bar{k}_1}$$



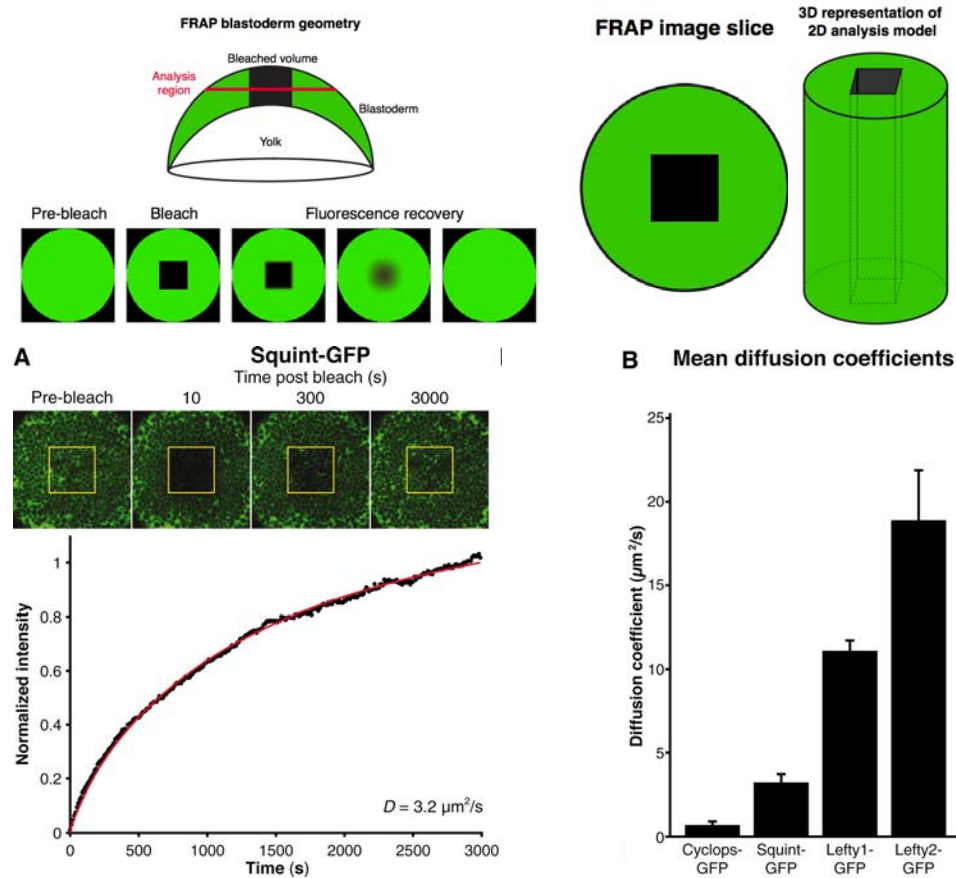
Measuring extracellular clearance rate constants with pulse-labelling assay



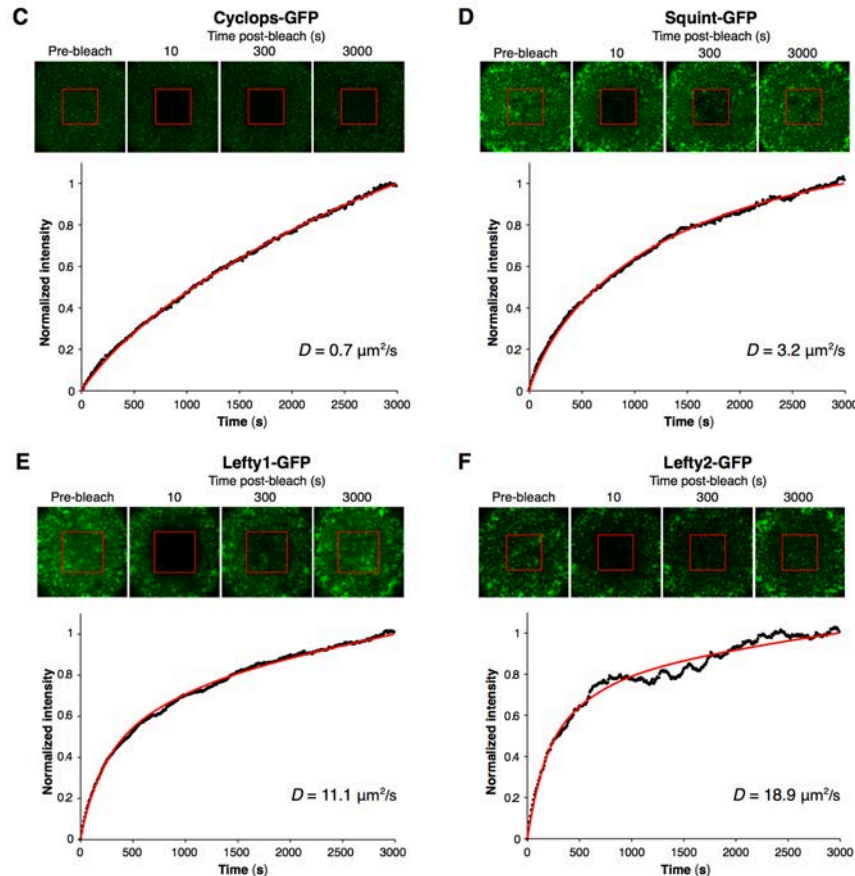
Protein half-lives increase only slightly as ranges increases

Differential clearance is a minor contributor to difference in Nodal and Lefty activity range

Measuring effective diffusion coefficients using Fluorescence Recovery After Photobleaching (FRAP)



Measuring effective diffusion coefficients using Fluorescence Recovery After Photobleaching (FRAP)



Increased protein diffusivity with increased ranges

Differential diffusivity is a major contributor to difference in Nodal and Lefty activity range

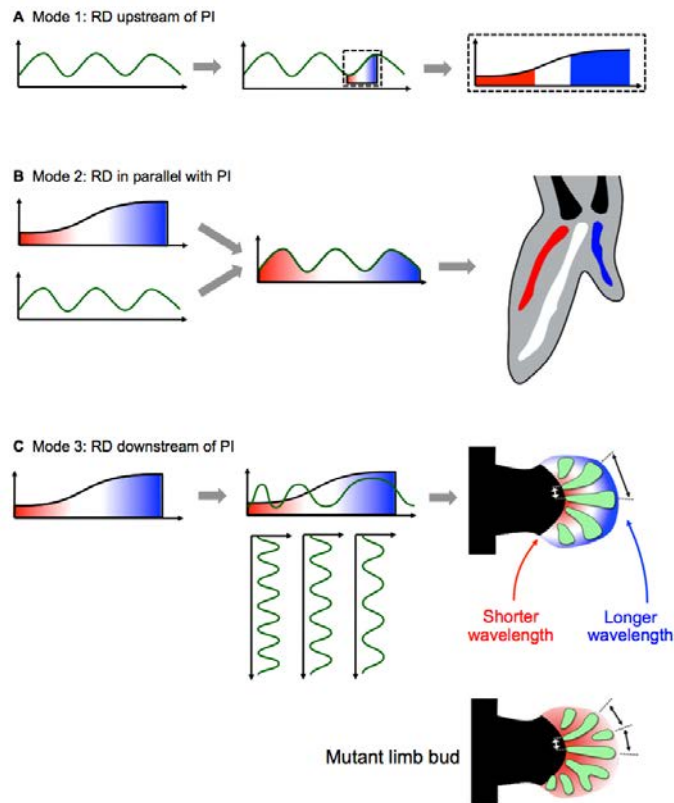
Conclusion

Differential diffusivity underlies differences in activator/
inhibitor range

Similar trends in half-lives, but differences in diffusivity are
much more pronounced compared to differences in
clearance

Previously described network topology and biophysical
properties support **activator/inhibitor reaction-diffusion
model of morphogenesis**

Modes of possible reaction-diffusion (RD)/positional information (PI) interaction



Thank you!